

```
from google.colab import files
uploaded = files.upload()
```

Choose files training.160...moticon.csv
training.1600000.processed.noemoticon.csv(text/csv) - 238803811 bytes, last modified: 29/11/2025 - 100% done
Saving training.1600000.processed.noemoticon.csv to training.1600000.processed.noemoticon.csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from wordcloud import WordCloud
import seaborn as sns
import re
```

```
df = pd.read_csv('training.1600000.processed.noemoticon.csv', encoding='latin-1',header =None)
df.columns = ["sentiment","id", "date", "query", "user", "text"]
```

```
df.shape
```

(1600000, 8)

```
df['sentiment'] = df['sentiment'].map({0: 'negative', 4: 'positive'})
```

```
df['sentiment'].value_counts()
```

count	
sentiment	
negative	800000
positive	800000

dtype: int64

```
df['data'] = pd.to_datetime(df['date'])
```

/tmp/ipython-input-2838137911.py:1: FutureWarning: Parsed string "Mon Apr 06 22:19:45 PDT 2009" included an un-recognized timezo
df['data'] = pd.to_datetime(df['date'])

```
df.head(2)
```

	sentiment	id	date	query	user	text	data	clean_text	weekday	hour
0	negative	1467810369	2009-04-06 22:19:45	NO_QUERY	_TheSpecialOne_	@switchfoot http://twitpic.com/2y1zl - Awww, t...	2009-04-06 22:19:45	httpwtipiccomyzl awww thats a bummer you s...	Monday	22
1	negative	1467810370	2009-04-06 22:19:45	NO_QUERY	" " " "	is upset that he can't	2009-04-06 22:19:45	is upset that he	Monday	22

```
import re
def clean_text(text):
    text = str(text).lower()
    text = re.sub(r'http\s+', '',text) # this removes urls
    text = re.sub(r'@\w+', '',text) # this removes mentions
    text = re.sub(r'#\w+', '',text) # this removes hashtags
    text = re.sub(r'[^\A-Za-z\s]', '',text) # this removes special char (corrected regex from [A-Za-z\s] to [^\A-Za-z\s])
    text = text.lower()
    return text
df['clean_text'] = df['text'].apply(clean_text)
df.head()
```

	sentiment	id	date	query	user	text	data	clean_text
0	negative	1467810369	Mon Apr 06 22:19:45 PDT 2009	NO_QUERY	_TheSpecialOne_	@switchfoot http://twitpic.com/2y1zl - Awww, t...	2009-04-06 22:19:45	httpwtipiccomyzl awww thats a bummer you s...
1	negative	1467810672	Mon Apr 06 22:19:49 PDT 2009	NO_QUERY	scotthamilton	is upset that he can't update his Facebook by ...	2009-04-06 22:19:49	is upset that he cant update his facebook by t...
2	negative	1467810917	Mon Apr 06 22:19:53 PDT 2009	NO_QUERY	mattycus	@Kenichan I dived many times for the ball. Man...	2009-04-06 22:19:53	i dived many times for the ball managed to sa...
3	negative	1467811184	Mon Apr 06 22:19:57 PDT	NO_QUERY	ElleCTF	my whole body feels itchy and like its on fire	2009-04-06 22:19:57	my whole body feels itchy and like its on fire

df.tail(3)

	sentiment	id	date	query	user	text	data	clean_text	weekday	hour
1599997	positive	2193601991	2009-06-16 08:40:49	NO_QUERY	bpbabe	Are you ready for your MoJo Makeover? Ask me f...	2009-06-16 08:40:49	are you ready for your mojo makeover ask me fo...	Tuesday	8
1599998	positive	2193602064	2009-06-16 08:40:49	NO_QUERY	tinydiamondz	Happy 38th Birthday to my boo of alll time!!!	2009-06-16 08:40:49	happy th birthday to my boo of alll time	Tuesday	8

df['clean_text'].head(3)

clean_text

0 httpwtipiccomyzl awww thats a bummer you s...

1 is upset that he cant update his facebook by t...

2 i dived many times for the ball managed to sa...

dtype: object

```
from wordcloud import WordCloud
import matplotlib.pyplot as plt

positive_texts = df[df['sentiment'] == 4]['clean_text']

positive_words = " ".join(positive_texts)

wordcloud = WordCloud(width=1200, height=600, background_color='black').generate(positive_words)

plt.figure(figsize=(14, 7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("Word Cloud - Positive Sentiment", fontsize=20)
plt.show()
```

[illegible]

```
from wordcloud import WordCloud
import matplotlib.pyplot as plt

negative_texts = df[df['sentiment'] == 0]['clean_text']

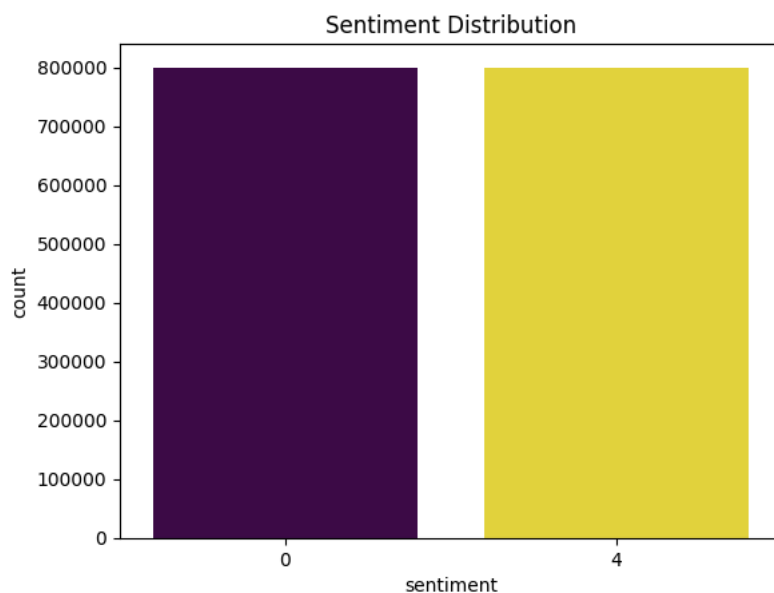
negative_words = " ".join(negative_texts)

wordcloud_neg = WordCloud(width=1200, height=600, background_color='black').generate(negative_words)

plt.figure(figsize=(14, 7))
plt.imshow(wordcloud_neg, interpolation='bilinear')
plt.axis('off')
plt.title("Word Cloud - Negative Sentiment", fontsize=20)
plt.show()
```

[illegible]

```
sns.countplot(data=df, x='sentiment', hue='sentiment', palette='viridis', legend=False);  
plt.title("Sentiment Distribution")  
plt.show()
```



```
from collections import Counter

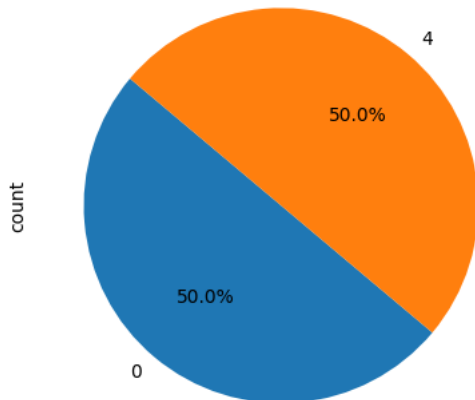
words = " ".join(df['clean_text']).split()
Counter(words).most_common(20)
```

```
[('i', 751301),
 ('to', 564591),
 ('the', 520094),
 ('a', 378001),
 ('my', 314119),
 ('and', 298466),
 ('you', 270155),
 ('is', 236031),
 ('it', 230924),
 ('for', 215724),
```

```
( 'in', 214386),
( 'of', 183447),
( 'im', 178401),
( 'on', 167122),
( 'me', 160210),
( 'so', 150834),
( 'have', 144550),
( 'that', 141356),
( 'but', 127522),
( 'just', 125070)]
```

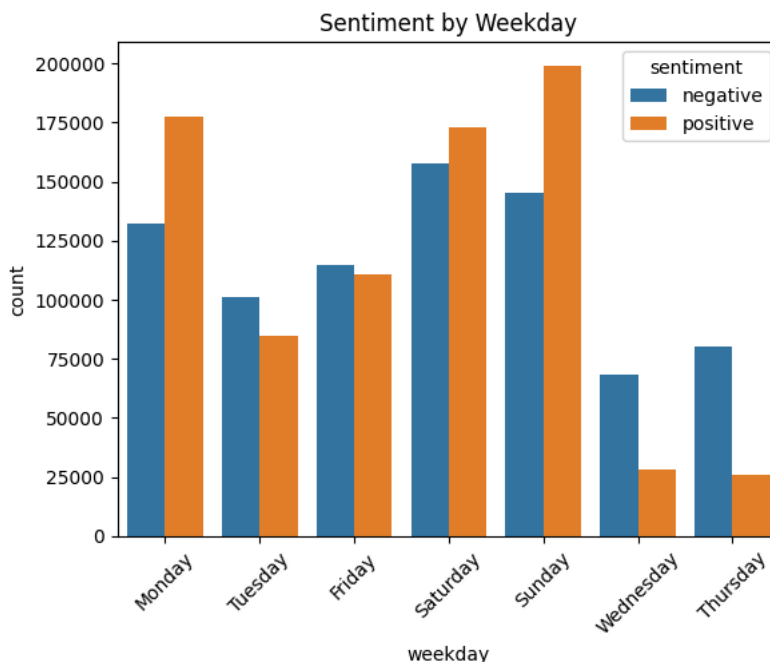
```
df['sentiment'].value_counts().plot(kind= 'pie', autopct='%1.1f%%', startangle=140 )
```

<Axes: ylabel='count'>



```
df['date'] = pd.to_datetime(df['date'], errors='coerce') # Convert 'date' column to datetime
df['weekday'] = df['date'].dt.day_name()
sns.countplot(data=df, x='weekday', hue='sentiment')
plt.xticks(rotation=45); plt.title("Sentiment by Weekday"); plt.show()
```

/tmp/ipython-input-3282396534.py:1: FutureWarning: Parsed string "Mon Apr 06 22:19:45 PDT 2009" included an un-recognized timezo
df['date'] = pd.to_datetime(df['date'], errors='coerce') # Convert 'date' column to datetime



```
df['date'] = pd.to_datetime(df['date'], errors='coerce')
twitter_sentiments = df.dropna(subset=['date']).copy()

sent_year = twitter_sentiments.groupby([twitter_sentiments['date'].dt.year, 'sentiment']).size().unstack(fill_value=0)
```

```

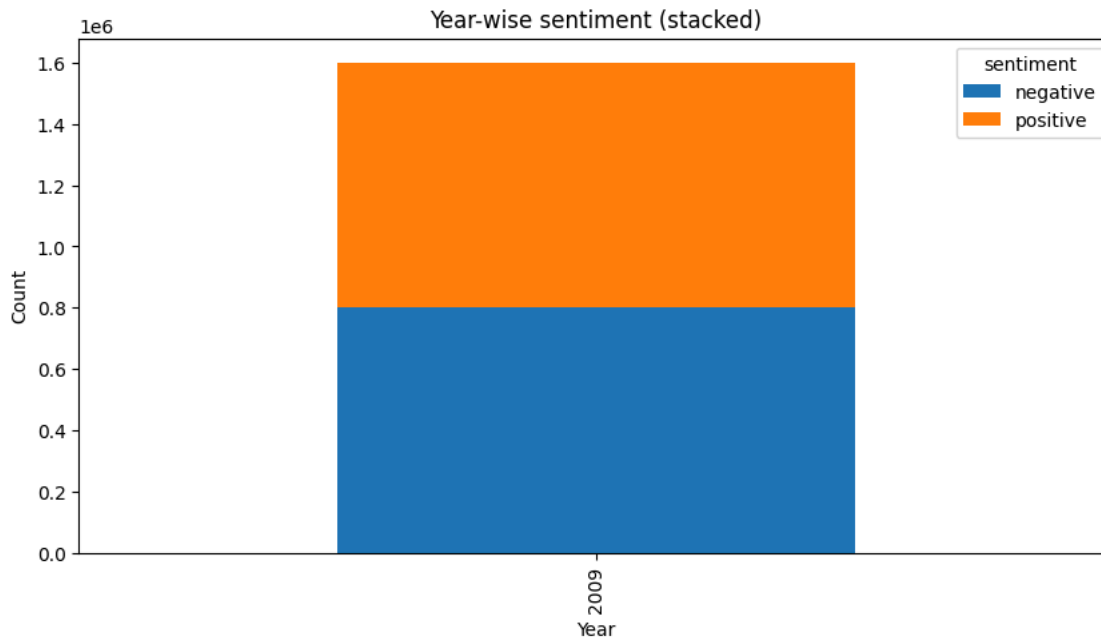
years = np.arange(sent_year.index.min(), sent_year.index.max()+1)
sent_year = sent_year.reindex(years, fill_value=0)

plt.figure(figsize=(10,5))
sent_year.plot(kind='bar', stacked=True, figsize=(10,5))
plt.title("Year-wise sentiment (stacked)")
plt.xlabel("Year"); plt.ylabel("Count")
plt.show()

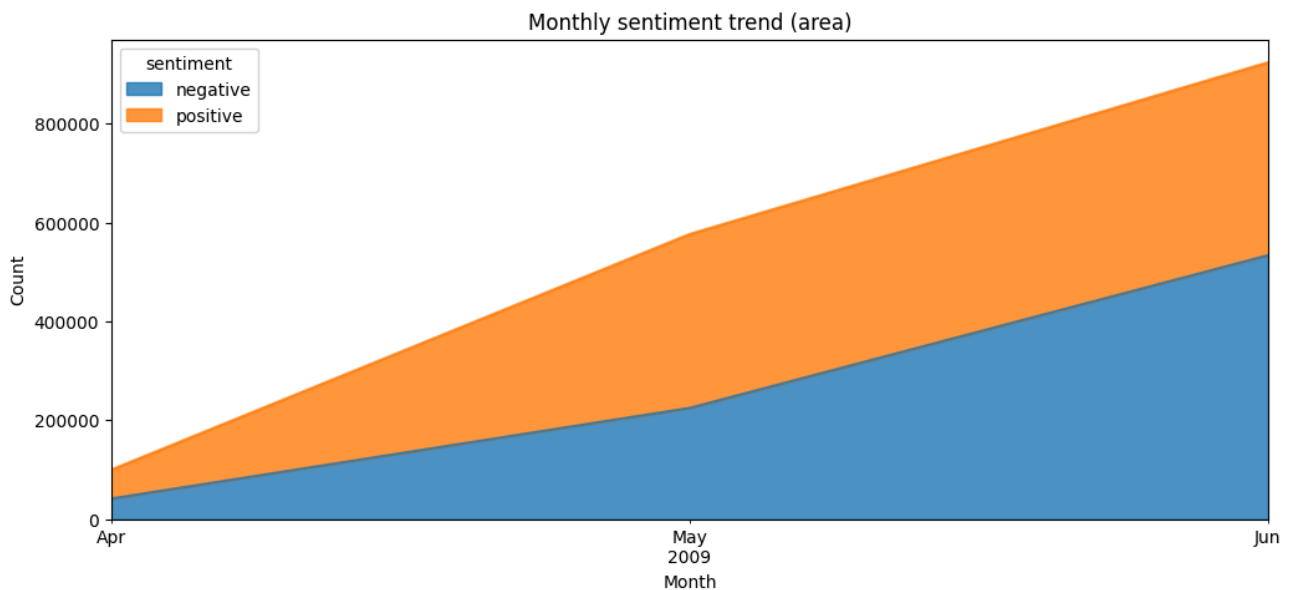
twitter_sentiments.set_index('date', inplace=True)
monthly = twitter_sentiments.groupby('sentiment').resample('M').size().unstack(0).fillna(0)
monthly.index = monthly.index.to_period('M').to_timestamp()
monthly.plot(kind='area', figsize=(12,5), alpha=0.8)
plt.title("Monthly sentiment trend (area)")
plt.xlabel("Month"); plt.ylabel("Count")
plt.show()

```

<Figure size 1000x500 with 0 Axes>



/tmp/ipython-input-2778578456.py:16: FutureWarning: 'M' is deprecated and will be removed in a future version, please use 'ME'
monthly = twitter_sentiments.groupby('sentiment').resample('M').size().unstack(0).fillna(0)



```

df['hour'] = df['date'].dt.hour
hour_sent = df.groupby(['hour', 'sentiment']).size().unstack(fill_value=0)

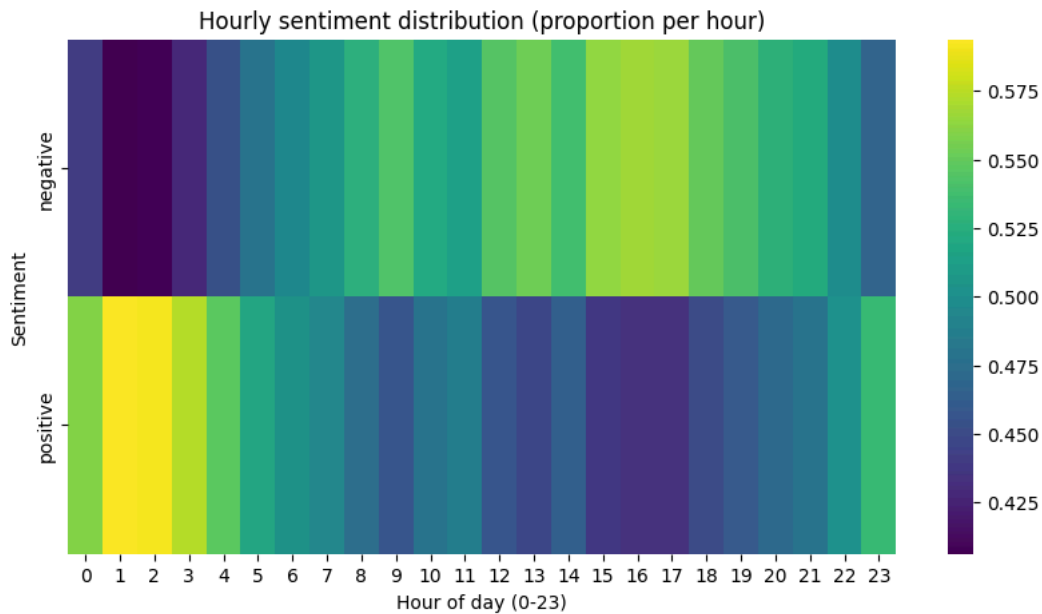
```

```

hour_pct = hour_sent.div(hour_sent.sum(axis=1), axis=0)

plt.figure(figsize=(10,5))
sns.heatmap(hour_pct.T, annot=False, cmap='viridis')
plt.title("Hourly sentiment distribution (proportion per hour)")
plt.xlabel("Hour of day (0-23)"); plt.ylabel("Sentiment")
plt.show()

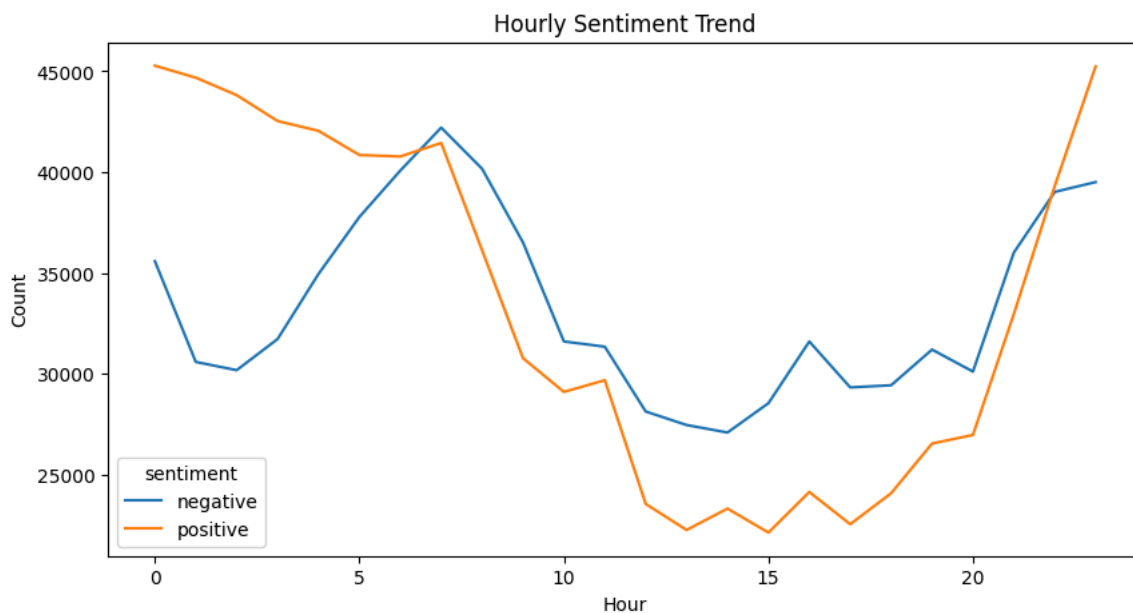
```



```

hourly = df.groupby([twitter_sentiments.index.hour, 'sentiment']).size().unstack()
hourly.plot(figsize=(10,5))
plt.title("Hourly Sentiment Trend"); plt.xlabel("Hour"); plt.ylabel("Count")
plt.show()

```



```

monthly_pct = monthly.div(monthly.sum(axis=1), axis=0)
monthly_pct.plot(kind='area', figsize=(12,5))
plt.title("Proportion of Sentiments Over Time")
plt.show()

```

